

Winter Examination – 2022

Course: B. Tech. Branch: Artificial Intelligence & Data Science Semester: III

Subject Code & Name: BTAIC303 – Data Structure and Algorithm using Python

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Explain the use of break and continue statement with suitable example.	(BT2/CO1)	6
B) What is list? Explain any 5 operation on list with suitable example.	(BT1/CO1)	6
C) Write a program to take one string as input and print total number of words in given string.	(BT3/CO1)	6
Q.2 Solve Any Two of the following.		12
A) What is class and object? Write a program to initialize and display details of two student using classes and objects?	(BT3/CO2)	6
B) What is Exception Handling? Explain how to handle exceptions?	(BT3/CO2)	6
C) Write a code to define a function which will take list as parameter and return the count of even number in list. Call function with appropriate parameter.	(BT3/CO2)	6
Q. 3 Solve Any Two of the following.		12
A) Write algorithms for deleting nodes from singly linked list.	(BT3/CO3)	6
B) Differentiate between array and linked list? Explain the node structure for singly linked list and doubly linked list with suitable example.	(BT2/CO3)	6
C) Write a program to perform push(), pop(), isEmpty(), isFull() and display() operations on stack.	(BT3/CO1)	6
Q.4 Solve Any Two of the following.		12
A) What is BST? Create a BST for 60, 56, 40, 34, 70, 80, 50, 45. After creation of BST perform deletion on node 34.	(BT3/CO4)	6
B) Define Hashing? Explain collision with suitable example?	(BT2/CO4)	6
C) Define Tree Data Structure? Explain several Tree Terminologies?	(BT3/CO4)	6
Q. 5 Solve Any Two of the following.		12
A) Write algorithm for insertion sort? Sort following element using insertion sort 40,30,20,50,45,60,10.	(BT3/CO5)	6
B) What is an algorithm? Explain time complexity and space complexity?	(BT4/CO5)	6
C) Write python program for Binary Search algorithm.	(BT3/CO5)	6